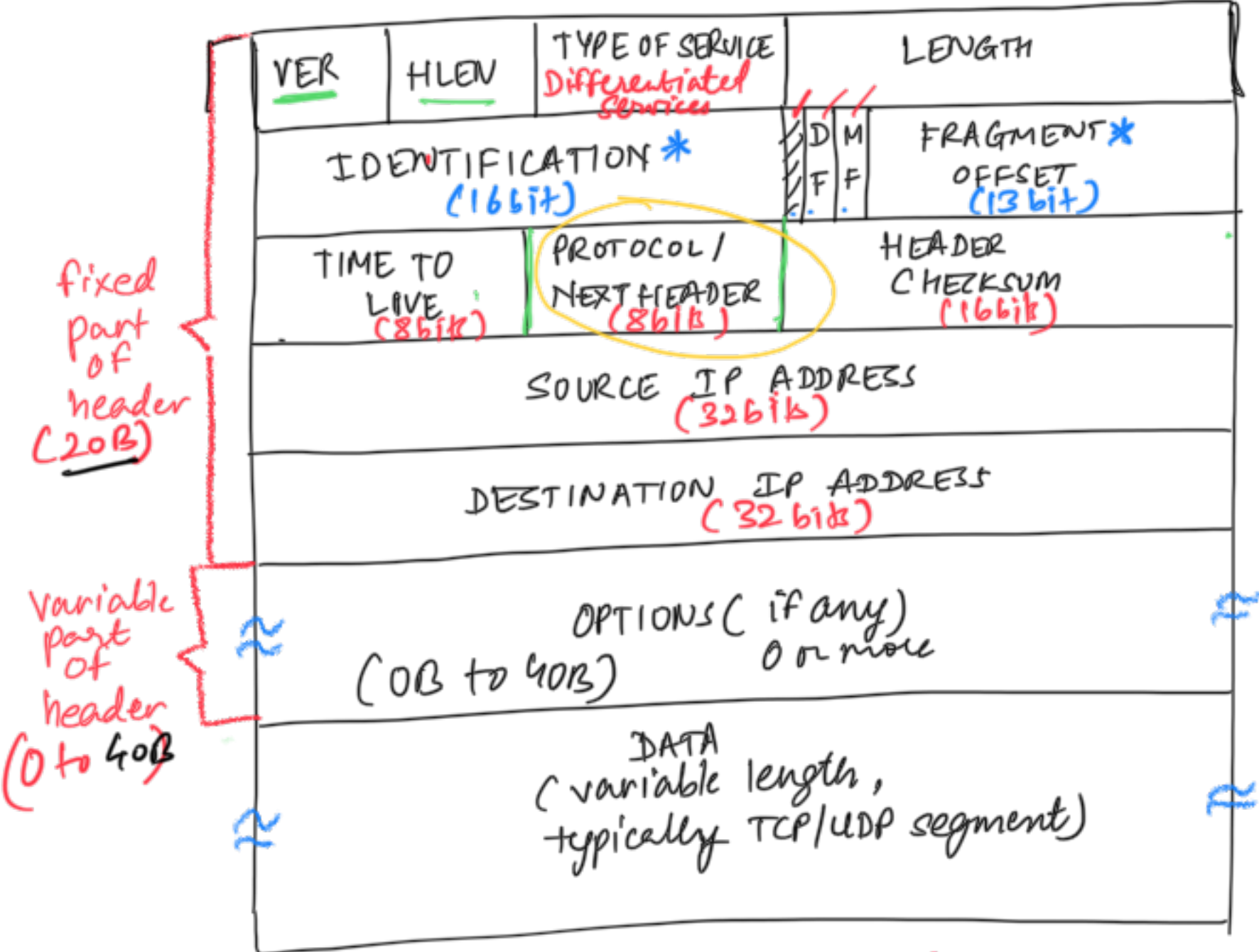
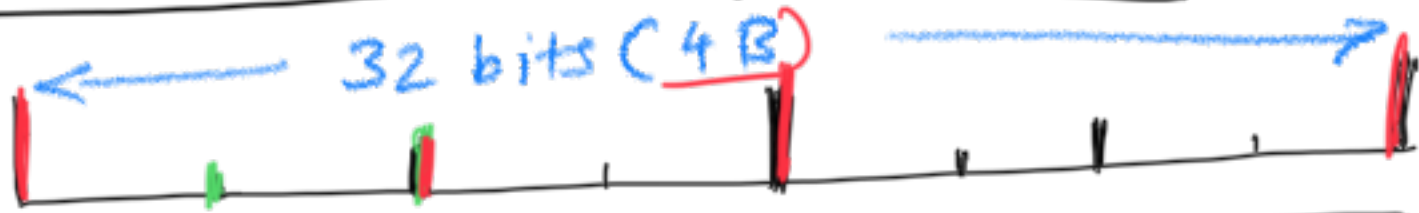


Lec 23 - IPv4 header + Routing in the Internet

7/11/23



* VER (4bits) : IP version 0100 $\Rightarrow$ IPV4

* HLEN (4bits) : Header length in words (1 word = 4B)

HLEN_{min} = 0101 = 5 * 4 = 20B

HLEN_{max} = 1111 = 15 * 4 = 60B

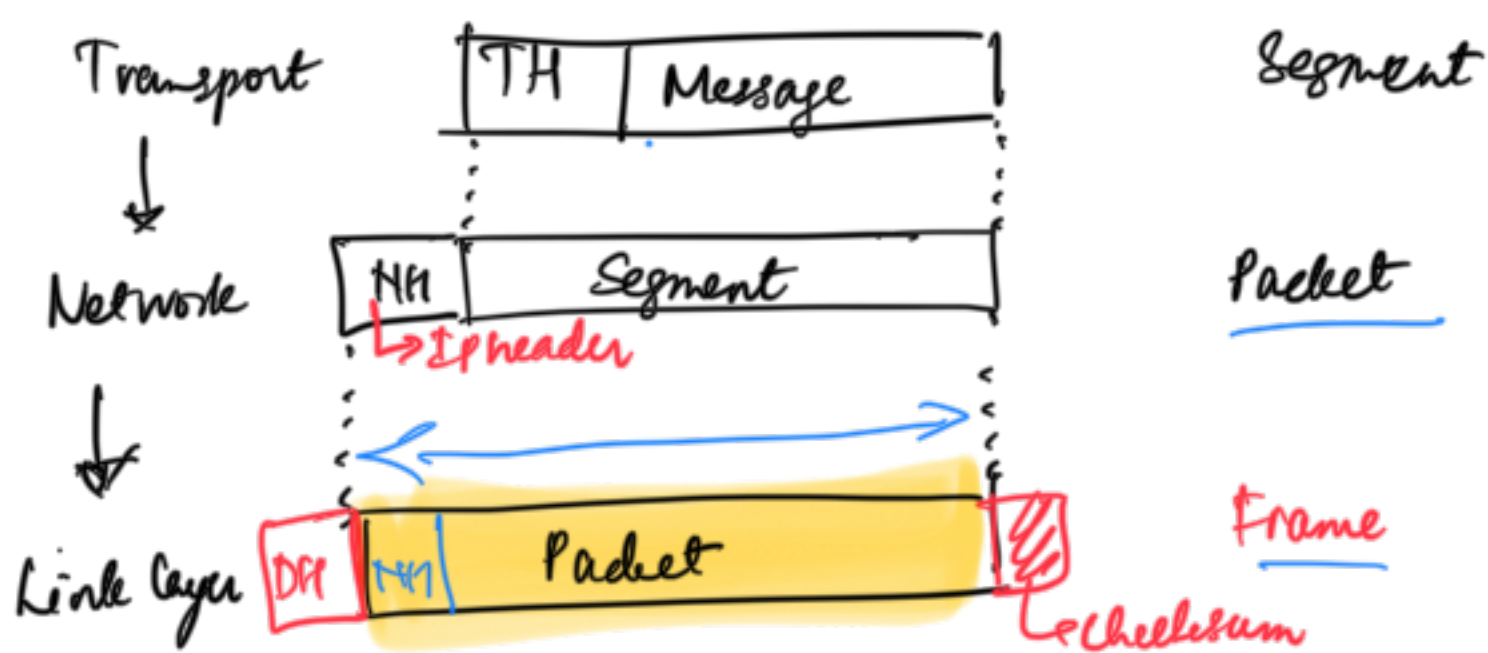
* TOS (8bits) : 3 bits : Priority (low) to 111 (high) $\rightarrow$ network control pkt

3 flags : Delay = 1 $\Rightarrow$ low delay

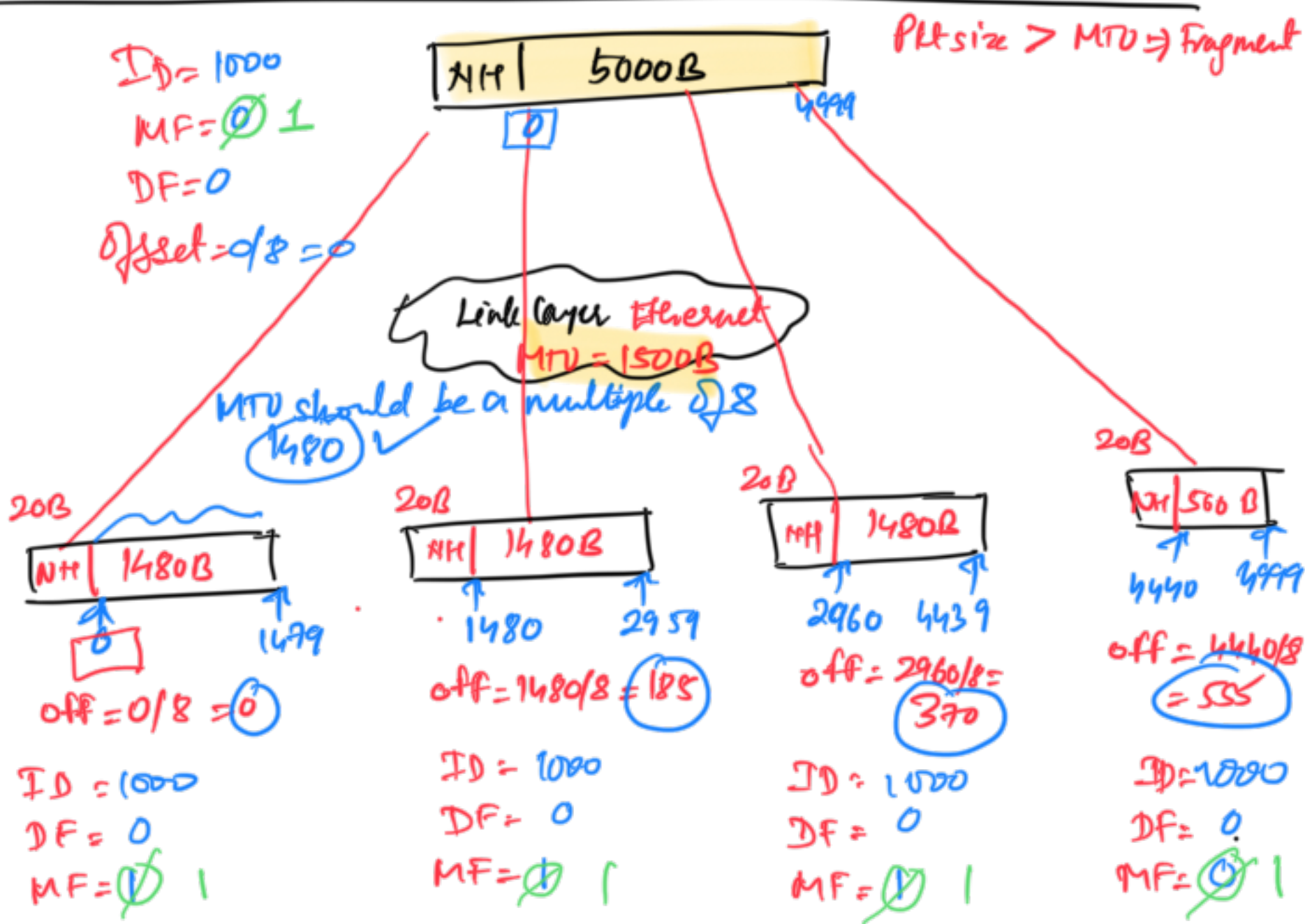
Throughput = 1 $\Rightarrow$ high

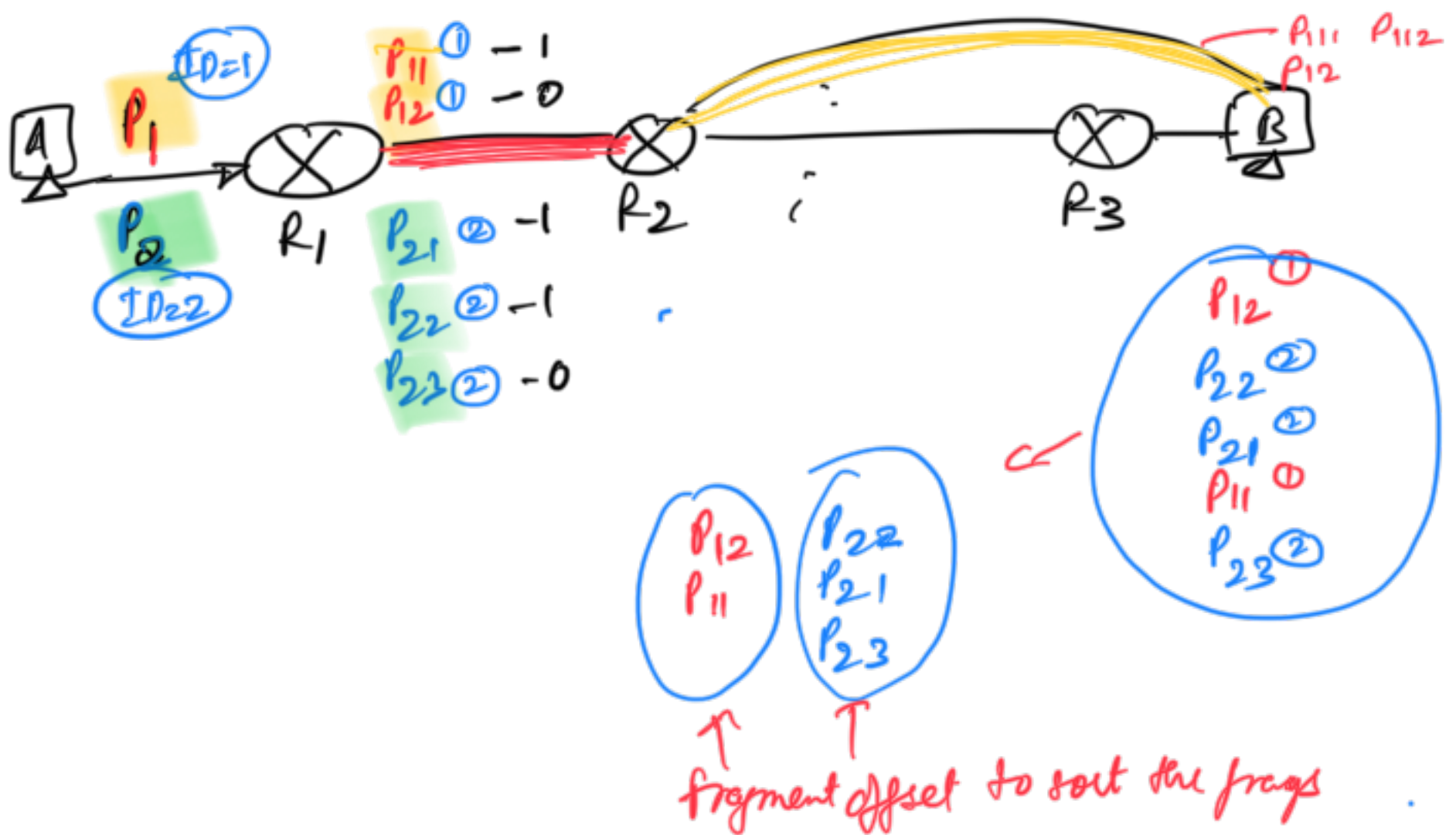
Reliability = 1 $\Rightarrow$ high

* length (16bits) : 2bits : unused (reserved) length of header + data in bytes
Max $2^{16} - 1$



$MTU = 1500B \Rightarrow \text{Ethernet}$
Maximum transfer unit
 $MTU = 1480$ excluding IP headers
 $(1500 - 20)$





* **DF** : Don't fragment
(1 bit)

DF = 1 $\Rightarrow$ Don't fragment the pkt

* **MF** : More fragments
(1 bit)

MF = 1 $\Rightarrow$ More frags coming

MF = 0 $\Rightarrow$ last fragment

* **Identification** : used to identify fragments of a single pkt
(16 bits)
 $\hookrightarrow$ ID for fragments of same pkt is
EQUAL

◦ Length: len (header + data) in bytes
(16b) $HLEN = 5 \text{ words} = 5 \times 4 = 20B$

data len = Length $= HLEN \times 4$
eg. $HLEN = 4000, HLEN = 5$
 $DATA LEN = 4000 - 5 \times 4 = 3980B$

◦ Identification: all fragments of same packet have same ID; used to identify fragments of same packet at Rx
(16b)

◦ Flags: DF: Don't Fragment
 $DF = 0$... do not fragment
 $DF = 1$... may fragment

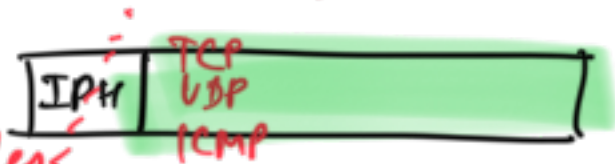
MF: More fragments
 $MF = 1$... more fragments
 $MF = 0$... last fragment

◦ Fragment offset: specifies the offset of fragment in the original packet
(13b) (used for sorting fragments at the Rx)

* TTL (Time to live): prevent pkts to roam in the network forever
(8bits)

• Src sets TTL
• each router decrements TTL by 1 or multiple
• $TTL = 0 \Rightarrow$ router drops pkt.

* Protocol (Next header):
(8bits)

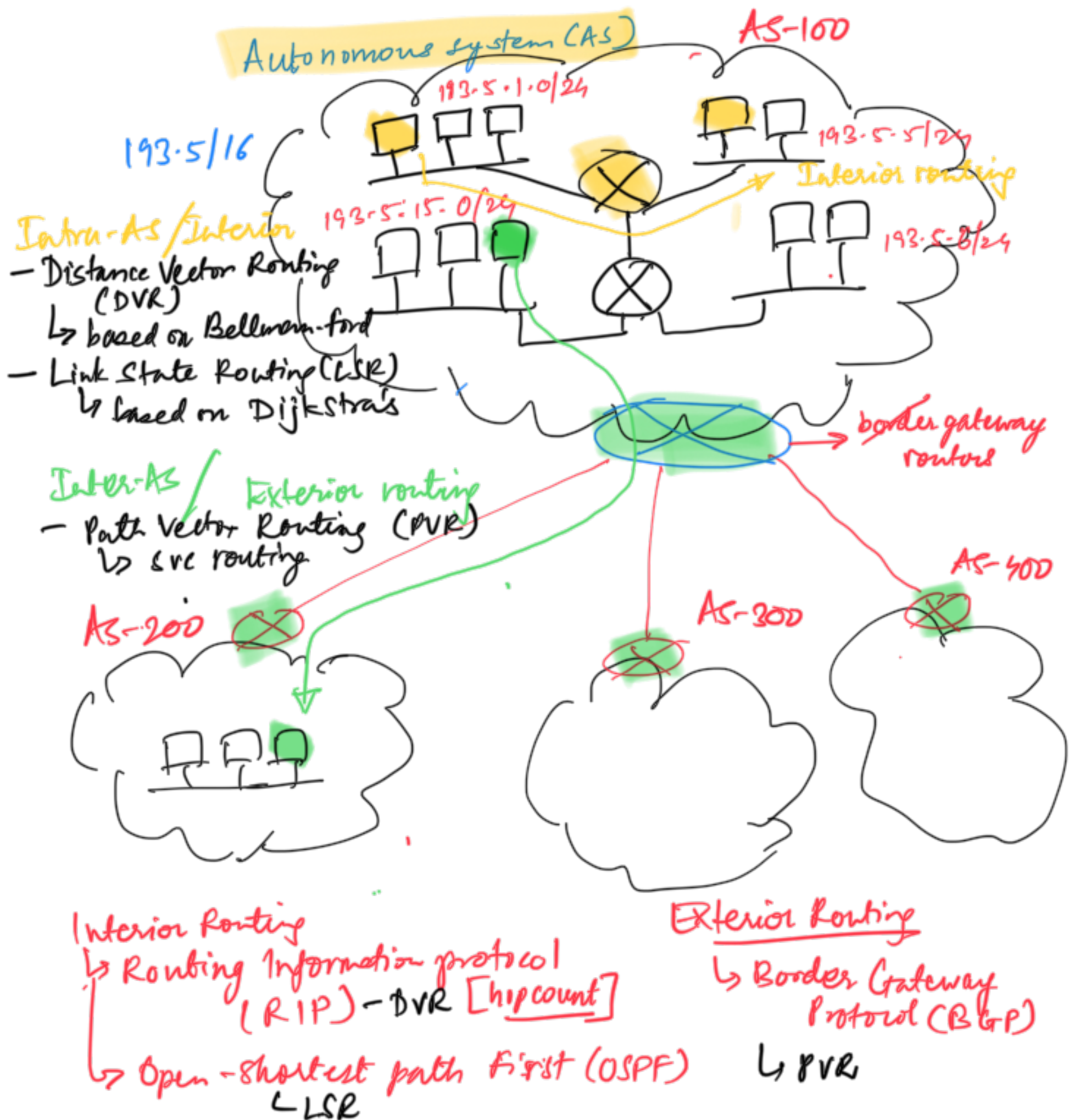


- specifies the header after IP header
- Protocol = 1: ICMP, 2: IGMP, 6: TCP, 17: UDP,

* Header checksum: checksum only for header
(16bits)
↳ error detection

Routing in the Internet

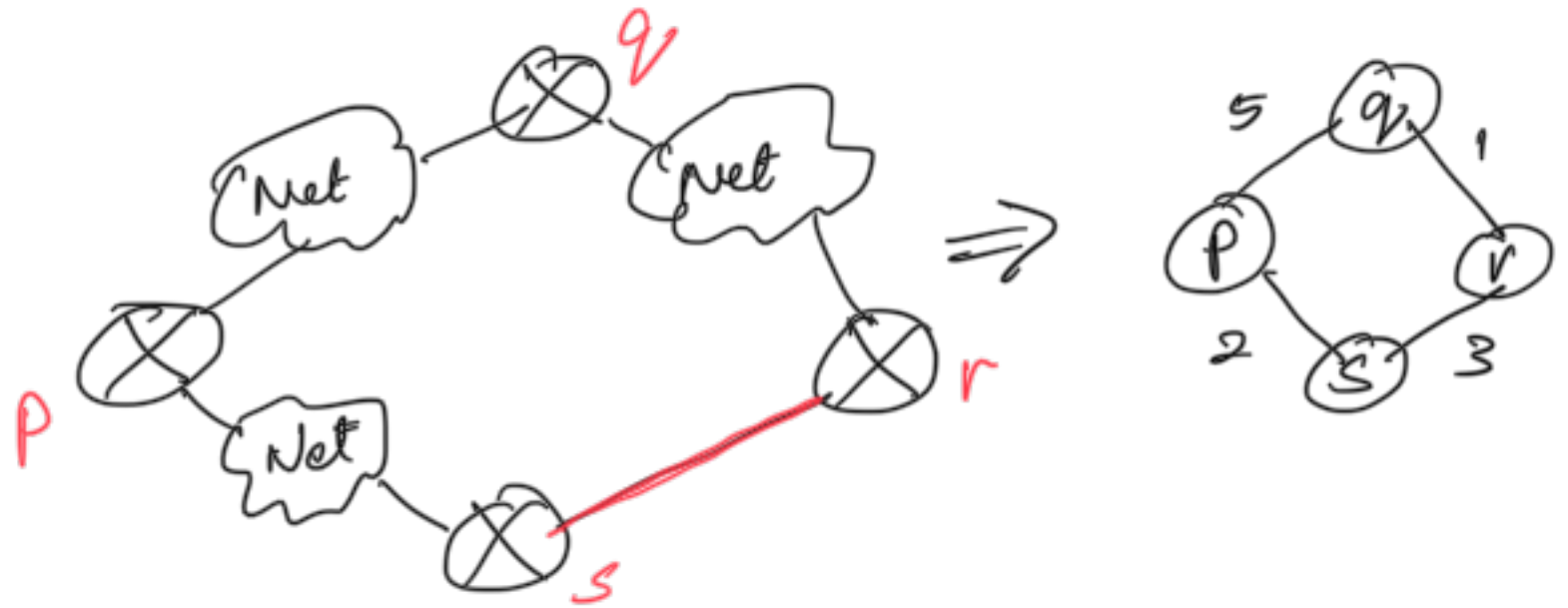
Routing: Select the best path to the destination
↳ delay, BW, distance, cost, ...



* Classification of Routing algorithms.

- Static $\Rightarrow$ Routing table entries are mostly fixed, (do not frequently)
 - Dynamic $\Rightarrow$ RT entries frequently change based on ongoing traffic, congested links, router/link failures, ...
-

* Dynamic Routing algorithms [INTERIOR ROUTING]



o Representation

Graph, $G = (N, E)$

vertices $\leftarrow$

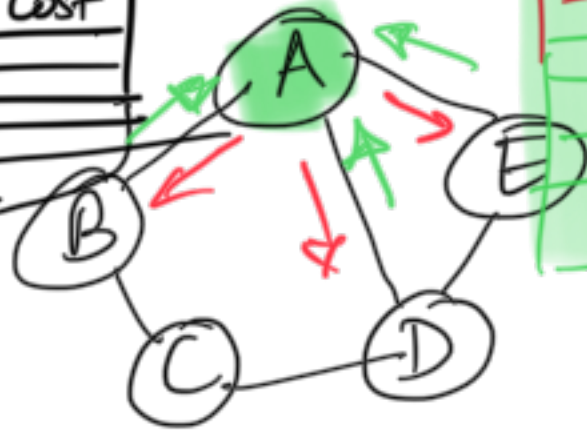
$\rightarrow$ networks, point-to-point links

* Distance Vector Routing (DVR) vs. Link State Routing (LSR)

Dynamic Routing
Interior Routing

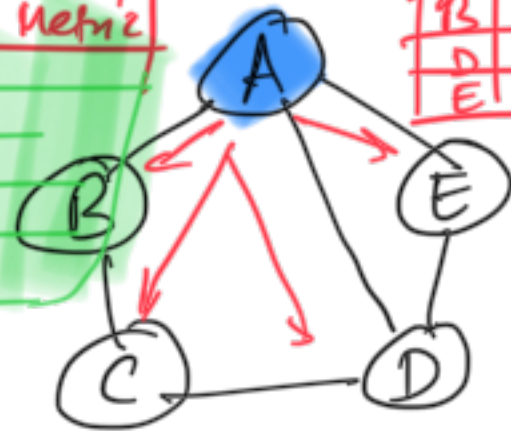
Distance Vector (DV)

Net	Cost
A	
B	
C	
D	
E	



Network	Next hop	Metric
A		
B		
C		
D		
E		

Neighbor		
B		
D		
E		



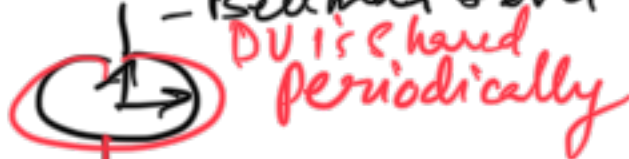
DVR

- RT initialization (Self cost = 0)
Neighbor cost = ∞
Rest = ∞

has entries for all nodes in network

- DV is shared only with the neighbors

- new RT is constructed - Bellman Ford



RIP: 30 sec

LSR

- RT initialization
- Every router shares neighbors RT info with entire network
- new RT is constructed - Dijkstra's
 - if RT entry has changed
 - if neighbor link fails / change in cost